OCEAN SCIENCE

Olympiad Study Guide

Grades 3 – 4 Advanced Level

- Ocean Chemistry
- Currents & Circulation
- Ocean Zones
- Bioluminescence
- Tides & Gravity
- Food Webs
- Climate & Carbon
- Plate Tectonics

Ocean Chemistry

Chem

SALINITY

Average ocean salinity: **3.5%** (35 parts per thousand)

- Salinity is measured in parts per thousand (‰).
- Dead Sea has ~34% salinity — 10× saltier than the ocean!
- Evaporation increases salinity; rain & rivers decrease it.
- Salt comes from erosion of rocks on land over millions of years.
- Polar seas are less salty because melting ice adds fresh water.

pH & DISSOLVED GASES

Ocean pH (average)

~8.1 (slightly alkaline)

CO_2 absorbed by ocean

~30% of all human emissions

Ocean acidification

pH dropped from 8.25 → 8.1 since 1750

Dissolved oxygen

Required by all marine animals to breathe

Thermocline

Layer where temperature drops sharply with depth

🌀 Ocean Currents & Circulation

THERMOHALINE CIRCULATION — The Ocean's Conveyor Belt

Cold, salty water is DENSER and sinks. Warm water rises. This difference in temperature (thermo) and salt (haline) drives a massive global circulation system that moves water around the entire planet — taking ~1,000 years to complete one cycle.

SURFACE CURRENTS

- Driven by WIND and Earth's rotation (Coriolis Effect)
- Flow clockwise in Northern Hemisphere, counter-clockwise in Southern
- Gulf Stream: carries warm water from Gulf of Mexico to Europe
- Speed: up to 9 km/h — much faster than deep currents
- Affect climate of nearby continents significantly

DEEP CURRENTS

- Driven by DENSITY differences (temperature + salinity)
- Cold dense water sinks in polar regions, travels along ocean floor
- Speed: very slow — about 1 cm per second
- Transport nutrients from deep to surface (upwelling)
- Critical for distributing heat and oxygen globally

UPWELLING

- Deep, cold, nutrient-rich water rises to surface
- Fuels phytoplankton blooms → entire food webs
- West coasts of continents have strong upwelling zones
- Peru's upwelling makes it one of the richest fisheries on Earth
- El Niño disrupts upwelling, causing ocean ecosystem collapse

🌊 Ocean Zones — Advanced

Zones

EPIPELAGIC

Sunlight Zone • 0–200 m

☀ **Full sunlight | Temp: ~20°C | Pressure: 1–20 atm**

Photosynthesis occurs; phytoplankton, zooplankton, tuna, sea turtles, dolphins

★ *Produces ~50% of Earth's oxygen via phytoplankton.*

MESOPELAGIC

Twilight Zone • 200–1,000 m

🌑 **Faint blue | Temp: ~5°C | Pressure: ~100 atm**

Bioluminescent fish, lanternfish, swordfish hunt here at night

★ *Many animals migrate UP at night to feed, then sink by day.*

BATHYPELAGIC

Midnight Zone • 1,000–4,000 m

⚫ **Zero light | Temp: ~4°C | Pressure: ~400 atm**

Anglerfish, viperfish, giant squid; all use bioluminescence

★ *Pressure here is 400× greater than at sea level.*

ABYSSOPELAGIC

Abyss • 4,000–6,000 m

⚫ **Zero light | Temp: ~2°C | Pressure: ~600 atm**

Sea cucumbers, brittle stars, bacteria on marine snow

★ *Marine snow: dead organic matter drifts down like snowflakes.*

HADALPELAGIC

Hadal Zone / Trenches • 6,000–11,000 m

⚫ **Zero light | Temp: ~1–2°C | Pressure: ~1,000+ atm**

Amphipods, snailfish — some found at 8,000 m depths

★ *Mariana Trench: deepest point on Earth at ~11,034 m.*

✨ Bioluminescence

Bio

Bioluminescence is the ability of living organisms to produce and emit LIGHT through a chemical reaction.

THE CHEMISTRY

Luciferin:	The light-producing molecule (the "fuel")
Luciferase:	The enzyme that triggers the reaction
Oxygen:	Required — luciferin + O_2 + luciferase = light
Color:	Usually blue-green (blue travels farthest in water)
No heat:	Called "cold light" — almost no energy is wasted as heat

WHY DO THEY GLOW?

Attract prey:	Anglerfish dangles glowing lure to lure fish close
Scare predators:	Burst of light startles or blinds attackers
Camouflage:	Counterillumination — match ocean light from above
Communication:	Firefly squid use light patterns to signal mates
Distraction:	Some eject glowing fluid to confuse predators while escaping

76% of deep-sea animals are bioluminescent • Found in: jellyfish, squid, fish, plankton, bacteria, worms, sharks • Deep sea: the most common form of communication on Earth

🌙 Tides & Gravity

WHAT CAUSES TIDES?

Tides are caused by the GRAVITATIONAL PULL of the Moon (and to a lesser extent, the Sun) on Earth's oceans. The Moon's gravity pulls the ocean toward it, creating a bulge of water. A second bulge forms on the opposite side due to inertia. As Earth rotates, these bulges create the rise and fall of tides.

○● SPRING TIDES

Occur during NEW and FULL moons. Sun, Moon, and Earth align. Gravitational forces ADD together → highest high tides and lowest low tides.

◐◑ NEAP TIDES

Occur during QUARTER moons. Sun and Moon are at 90°. Forces partially cancel → smaller tidal range (less difference between high and low tides).

📏 TIDAL RANGE

Bay of Fundy, Canada has the world's highest tidal range: ~16 meters (53 feet)! Most places have 1–3 meter ranges.

🦀 TIDAL ZONES

Intertidal zone: exposed at low tide, submerged at high tide. Organisms must survive dramatic changes in moisture, temperature, and salinity.

Ocean Food Webs & Trophic Levels

Food

TROPHIC PYRAMID

Level 5 — Apex Predators

Orca, Great White Shark

Level 4 — Tertiary Consumers

Tuna, Seal, Barracuda

Level 3 — Secondary Consumers

Squid, Mackerel, Jellyfish

Level 2 — Primary Consumers

Zooplankton, Krill, Small Fish

Level 1 — Producers

Phytoplankton, Seaweed, Kelp

KEY CONCEPTS

10% Rule:

Only ~10% of energy transfers up each trophic level. 90% is lost as heat, movement, and waste.

Keystone Species:

A species that has a disproportionately large effect on its ecosystem. Sea otters control sea urchin populations → protect kelp forests.

Marine Snow:

Dead organisms and feces fall slowly to the ocean floor, feeding deep-sea communities — a key energy pathway.

Trophic Cascade:

Removing a top predator causes chain reactions — e.g., no sharks → more rays → fewer shellfish.

Phytoplankton:

The foundation of almost ALL ocean food webs; produce ~50% of Earth's oxygen via photosynthesis.

Coral Reefs — The Rainforests of the Sea

Coral

Coral reefs cover <1% of the ocean floor yet support ~25% of ALL marine species. Coral animals are called POLYPS and have a symbiotic relationship with algae called ZOOXANTHELLAE.

CORAL ANATOMY

- Coral POLYP: tiny animal with a soft body
- Hard calcium carbonate skeleton underneath
- Zooxanthellae algae live INSIDE polyp tissue
- Algae provide 90% of polyp's energy via photosynthesis
- Polyp provides shelter + CO_2 to algae → mutualism

CORAL BLEACHING

- Occurs when water gets too warm (>1°C above normal)
- Stress causes polyps to EXPEL their zooxanthellae
- Coral turns white (bleached) and may starve
- Can recover IF water cools quickly enough
- Great Barrier Reef: 50% bleached since 2016

REEF TYPES

- Fringing reefs: grow directly from shoreline
- Barrier reefs: separated from shore by lagoon
- Atolls: ring-shaped reefs over sunken volcanoes
- Great Barrier Reef: largest (~2,300 km, visible from space)
- Found in warm, shallow, clear tropical waters

Oceans & Climate Change

Climate

CARBON CYCLE

- Oceans absorb ~30% of CO_2 humans produce
- Dissolved CO_2 forms carbonic acid (H_2CO_3)
- Acidification dissolves shells of mollusks & coral
- Cold water absorbs more CO_2 than warm water
- Phytoplankton store carbon when they die and sink (biological pump)

HEAT ABSORPTION

- Oceans have absorbed ~90% of excess heat since 1970
- Ocean warming causes sea levels to rise (thermal expansion)
- Warmer oceans: more powerful hurricanes & cyclones
- Ice sheet melt adds fresh water, disrupting salinity & currents
- AMOC (Atlantic current) is slowing due to warming — threatens European climate

EL NIÑO & LA NIÑA

- El Niño: warm water spreads east across Pacific → droughts, floods worldwide
- La Niña: opposite effect — cooler water dominates
- Occur every 2–7 years; affect weather on every continent
- Named by Peruvian fishers (El Niño = 'The Boy' in Spanish, appears near Christmas)
- ENSO = El Niño Southern Oscillation — full cycle

Ocean Floor & Plate Tectonics

Geology

The ocean floor is as diverse as any landscape on land — with mountains, plains, canyons, and volcanoes. It is constantly being created and destroyed by tectonic plate movement.

Mid-Ocean Ridges

Underwater mountain chains where tectonic plates DIVERGE (move apart). Magma erupts and creates new ocean crust. The Mid-Atlantic Ridge is the longest mountain range on Earth (65,000 km).

Ocean Trenches

Formed where one plate SUBDUCTS (dives) under another. The Mariana Trench (~11,034 m) is the deepest point on Earth. Pacific Ring of Fire creates most trenches.

Hydrothermal Vents

Cracks in ocean floor where superheated water (up to 400°C) shoots out. Chemosynthetic bacteria at vents support entire ecosystems WITHOUT sunlight — key evidence for life on other planets.

Continental Shelf

Shallow, gently sloping underwater extension of continents (0–200 m deep). Most productive ocean zone — 90% of ocean fishing occurs here. Ends at the continental slope.

Abyssal Plain

Vast, flat ocean floor (3,000–6,000 m depth). Covered by fine sediment called ooze. Among the least explored regions on Earth.

Seamounts

Underwater volcanoes that don't reach the surface. Create hotspots for marine biodiversity. ~100,000 seamounts in world's oceans; few have been explored.

Deep-Sea Animal Adaptations

Adapt

Animals in the deep ocean must survive crushing pressure, total darkness, near-freezing temperatures, and scarce food. They have evolved extraordinary adaptations.

Anglerfish

- Bioluminescent lure on head to attract prey
- Huge mouth — can swallow prey larger than itself
- Extremely flexible stomach stretches to fit large meals

Giant Squid

- Largest eyes of any animal (~30 cm wide) to detect faint light
- Counter-shading: dark on top, light below for camouflage
- Chromatophores: can change skin color in milliseconds

Vampire Squid

- Not actually a squid — unique class Vampyromorphida!
- Flips cloak of webbing over body to appear larger
- Can survive in near-anoxic (very low oxygen) zones

Viperfish

- Hinged skull tilts back to fit large prey
- Longest teeth relative to body size of any fish
- Light organ along belly for counterillumination

Dumbo Octopus

- Ear-like fins for gentle locomotion at extreme depths
- Deepest-living octopus: found at 6,957 m depth!
- Swallows prey whole — no beak grinding at those pressures

Barreleye Fish

- Transparent head — brain is visible!
- Tubular eyes rotate to look UP through the head
- Spots bioluminescent prey silhouetted from above

Ocean Exploration & Key Scientists

Explore

1872–1876	**HMS Challenger Expedition** First scientific ocean survey; discovered ~4,700 new species; pioneered modern oceanography.
1960	**Trieste Dive to Mariana Trench** Jacques Piccard & Don Walsh first humans to reach deepest point (~10,916 m) in the bathyscaphe Trieste.
1977	**Hydrothermal Vents Discovered** Alvin submersible finds life at vents — revolutionized understanding of life on Earth (and possibly other planets).
1985	**RMS Titanic Found** Robert Ballard's team locates Titanic at ~3,800 m using ROVs — pioneering deep-sea robotic exploration.
2012	**James Cameron Solo Dive** Director Cameron reaches Challenger Deep alone in DEEPSEA CHALLENGER — collects samples and video.
2023	**OceanGate Titan Implosion** Submersible implodes at ~3,800 m visiting Titanic — highlights extreme dangers of deep-sea exploration.

⚡ Olympiad Fast Facts & Numbers to Know

Facts

Ocean area:	361 million km² (~71% of Earth's surface)
Average depth:	3,688 m (~12,100 ft)
Deepest point:	Challenger Deep, Mariana Trench: ~11,034 m
Tallest ocean mountain:	Mauna Kea (base to peak): 10,210 m — taller than Everest!
Ocean salinity:	Average 35 ‰ (3.5%)
Dead Sea salinity:	~340 ‰ (10× saltier than ocean)
Pacific size:	Larger than all land combined (~165 million km²)
Oxygen from ocean:	~50% of Earth's oxygen from phytoplankton

Bioluminescent organisms:	~76% of deep-sea animals
Explored ocean:	Only ~20% of the ocean has been mapped in detail
Mariana Trench pressure:	~1,086 atm (1,000× more than sea level)
Global conveyor belt:	~1,000 years for one complete cycle
Bay of Fundy tidal range:	~16 m — highest in the world
Speed of sound in ocean:	~1,500 m/s (5× faster than in air)
Ocean pH (current):	~8.1 (was 8.25 before industrial era)
Carbon absorbed by ocean:	~30% of human CO_2 emissions

📖 Advanced Vocabulary — Know These Cold

Vocab

Thermohaline
Related to both temperature (thermo) and salinity (haline) — drives deep ocean circulation

Bioluminescence
Light produced by living organisms through a chemical reaction involving luciferin

Upwelling
Deep cold nutrient-rich water rising to the surface, fueling marine productivity

Subduction
One tectonic plate diving under another, creating trenches

Ocean Acidification
Decrease in ocean pH caused by absorption of CO_2 from the atmosphere

ENSO
El Niño Southern Oscillation — cycle of warm/cool Pacific water affecting global weather

Chemosynthesis
Making food from chemicals (not sunlight) — used by bacteria at hydrothermal vents

Phytoplankton
Microscopic plant-like organisms in the ocean; base of food web + produce ~50% of Earth's O_2

Trophic Level
A feeding level in a food chain or food web

Coriolis Effect
Earth's rotation deflects moving objects/wind/ocean currents to the right (N) or left (S)

Zooxanthellae
Symbiotic algae living inside coral polyps; provide energy through photosynthesis

Hadal Zone
Ocean zone below 6,000 m; found only in deep ocean trenches